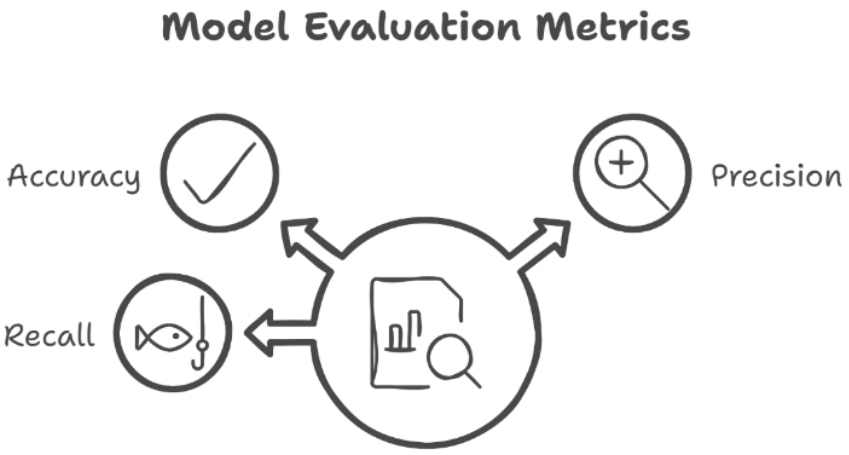


Why Model Evaluation Matters

Without robust evaluation, it is impossible to determine whether a model performs effectively or simply overfits the training data. Evaluation metrics provide quantitative measures of a model's predictive performance, enabling:

- **Comparison of Models:** Evaluate different algorithms to find the best-performing one.
- **Understanding Trade-offs:** Identify the balance between various metrics like precision and recall.
- **Guiding Model Improvement:** Highlight areas where the model needs refinement.

Common Metrics for Classification

1. Accuracy

Accuracy is the most straightforward metric and measures the proportion of correctly classified instances out of the total instances. It works well when the dataset has a balanced class distribution.

Formula:

Accuracy=Number of Correct PredictionsTotal Number of Predictions×100\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \times 100